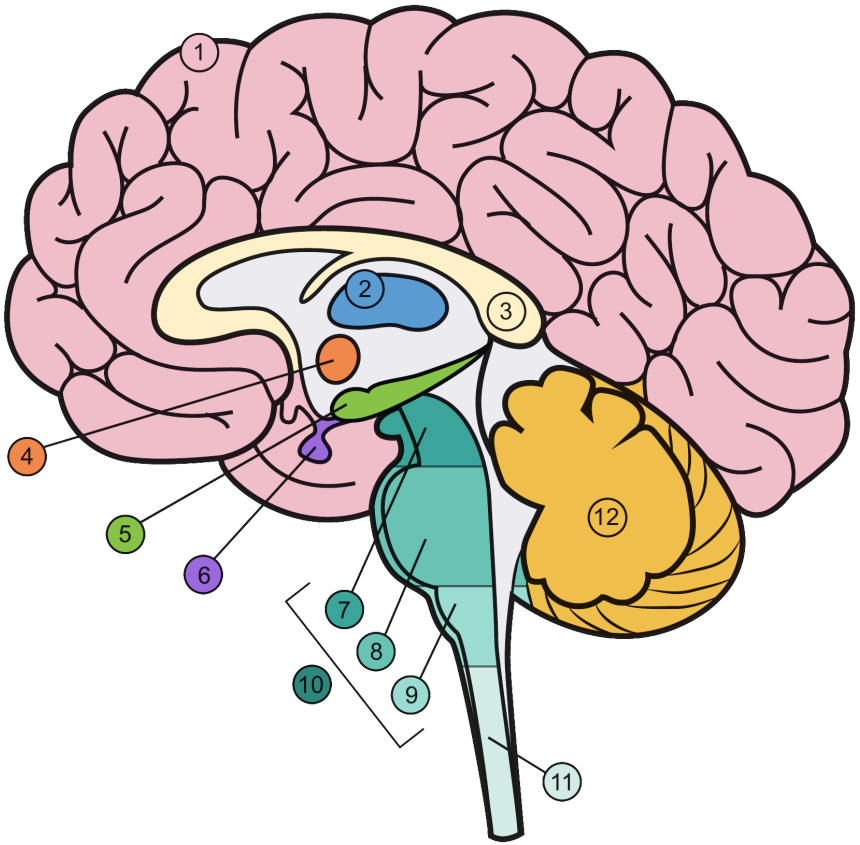
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Brain Anatomy — Reference Chart

MCB80x · midsagittal view · 12 structures, divisions, functions, and exam facts

DIAGRAM



Midsagittal section, anterior (front) to the left. Gray = ventricles / unlabeled tissue.

DEVELOPMENTAL DIVISIONS

Forebrain	Midbrain	Hindbrain	Other
Telencephalon: 1 3 5	Mesencephalon: 7	Pons 8 Medulla 9	Gland 6
Diencephalon: 2 4		Cerebellum 12	Cord 11

KEY

#	Structure	Function · exam fact
1	**Cerebral cortex** Telencephalon · gray matter	Six-layered outer sheet; perception, voluntary movement, language, planning. **FACT** Four lobes (frontal, parietal, temporal, occipital); gyri and sulci multiply surface area.
2	**Thalamus** Diencephalon	Relay hub sending every sense except smell on to the cortex. **FACT** LGN relays vision, MGN relays hearing.
3	**Corpus callosum** Telencephalon · white matter	~200 million axons linking left and right hemispheres. **FACT** Cut in "split-brain" surgery for severe epilepsy.
4	**Hypothalamus** Diencephalon	Homeostasis: temperature, hunger, thirst, circadian rhythm (SCN). **FACT** Makes oxytocin and ADH; releasing hormones steer the anterior pituitary.
5	**Hippocampus** Telencephalon · limbic	Forms new declarative and spatial memories; place cells. **FACT** Patient H.M.: bilateral removal → anterograde amnesia. LTP first shown here.
6	**Pituitary gland** Endocrine gland	Anterior lobe: GH, ACTH, TSH, LH/FSH, prolactin. **FACT** Posterior lobe is neural tissue — hypothalamic axon terminals releasing oxytocin and ADH.
7	**Midbrain** Mesencephalon	Colliculi: superior (visual orienting), inferior (auditory). **FACT** Substantia nigra dopamine loss → Parkinson's disease.
8	**Pons** Hindbrain · metencephalon	Relays cortex → cerebellum; helps set breathing rhythm and REM sleep. **FACT** Cranial nerves V–VIII exit at the pons and pontomedullary junction.
9	**Medulla oblongata** Hindbrain · myelencephalon	Heart rate, blood pressure, breathing, swallowing. **FACT** Corticospinal tracts cross here, so each hemisphere moves the opposite side.
10	**Brainstem** = 7 + 8 + 9	Links brain to spinal cord; reticular formation controls arousal. **FACT** Houses cranial nerve nuclei III–XII.
11	**Spinal cord** Central nervous system	Ascending sensory and descending motor tracts; spinal reflexes. **FACT** Dorsal root = sensory in, ventral root = motor out.
12	**Cerebellum** Hindbrain · metencephalon	Coordination, balance, timing, motor learning. **FACT** Holds ~80% of the brain's neurons; Purkinje cells are its sole output.

Personal MCB80x study chart — not an official HarvardX/edX publication. Line art: Ask A Biologist (askabiologist.asu.edu), CC BY-NC-SA; colorized.

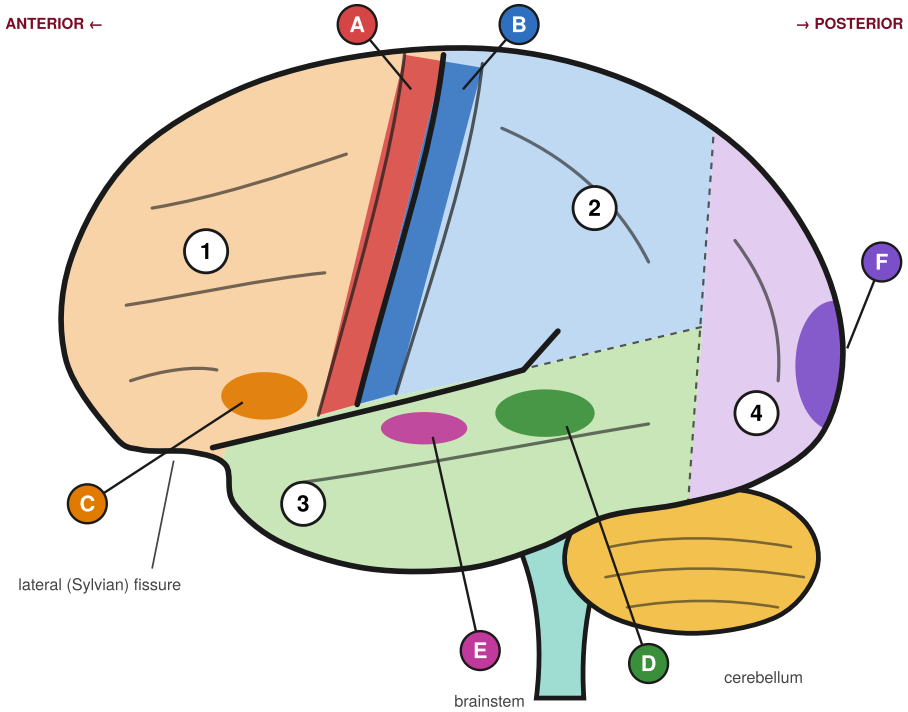
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Cerebral Cortex — Lobes & Functional Areas

MCB80x · lateral view · 4 lobes and 6 key cortical areas

DIAGRAM — LEFT HEMISPHERE, LATERAL VIEW



Schematic. Dashed lines are imaginary lobe borders; area sizes and positions are approximate.

Landmarks to know

- **Central sulcus** splits frontal from parietal — motor (A) in front, sensory (B) behind.
- **Lateral fissure** separates the temporal lobe from the frontal and parietal lobes above it.
- Language loop: **Wernicke (D) → arcuate fasciculus → Broca (C)**; usually left hemisphere.

KEY

#	Structure	Function · exam fact
1	**Frontal lobe** Anterior to central sulcus	Planning, decisions, working memory, personality (prefrontal); voluntary movement. **FACT** Phineas Gage: prefrontal damage changed personality and impulse control.
2	**Parietal lobe** Central sulcus → occipital	Touch, body position, spatial attention, combining the senses. **FACT** Right parietal damage → hemispatial neglect of the left side of space.
3	**Temporal lobe** Below lateral fissure	Hearing, language comprehension, object and face recognition; memory (medial). **FACT** Fusiform face area damage → prosopagnosia (face blindness).
4	**Occipital lobe** Posterior pole	Vision: color, motion, form processed in separate areas. **FACT** Each hemisphere sees the opposite visual field.
A	**Primary motor cortex** M1 · precentral gyrus	Sends commands for voluntary movement to the opposite side of the body. **FACT** Motor homunculus: hands and face get oversized territory.
B	**Primary somatosensory** S1 · postcentral gyrus	Touch, pain, temperature, and limb position from the opposite side. **FACT** Sensory homunculus mapped by Penfield's stimulation studies.
C	**Broca's area** Left inferior frontal gyrus	Speech production and grammar. **FACT** Broca's aphasia: slow, effortful speech; comprehension mostly spared.
D	**Wernicke's area** Left posterior sup. temporal	Language comprehension. **FACT** Wernicke's aphasia: fluent but meaningless speech, poor comprehension.
E	**Primary auditory cortex** A1 · Heschl's gyrus	First cortical stop for sound; mostly tucked inside the lateral fissure. **FACT** Tonotopic: neurons arranged by sound frequency.
F	**Primary visual cortex** V1 · calcarine sulcus	First cortical stop for vision; mostly on the medial surface. **FACT** Retinotopic map; input arrives from the thalamus's LGN.

Personal MCB80x study chart — not an official HarvardX/edX publication.

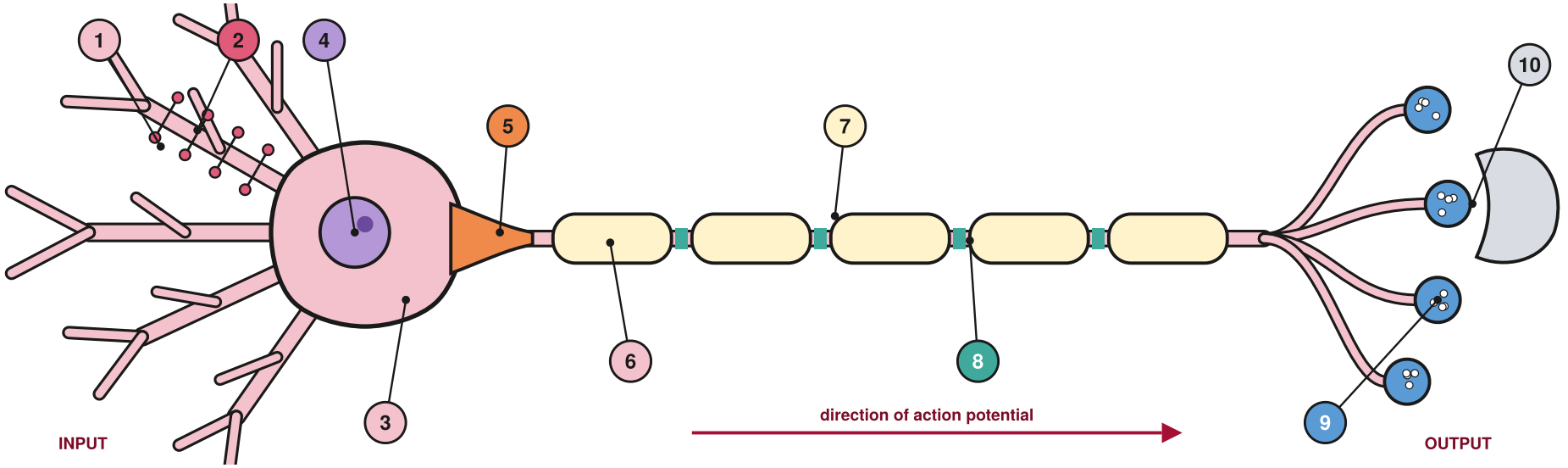
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Neuron Anatomy — Parts & Roles

MCB80x · input → integration → conduction → output

DIAGRAM — MYELINATED NEURON



Schematic, not to scale. Signal flows left to right: dendrites → soma → axon → terminals.

#	Structure	Function · exam fact
1	**Dendrites** Input zone	Branches that receive signals from other neurons. **FACT** Graded EPSPs/IPSPs spread passively and shrink with distance.
2	**Dendritic spines** Input zone	Tiny protrusions holding most excitatory synapses. **FACT** Spines grow and reshape with learning (LTP).
3	**Soma (cell body)** Integration	Holds the nucleus and protein-making machinery; sums incoming inputs. **FACT** Spatial + temporal summation decide whether to fire.
4	**Nucleus** Integration	Stores DNA; controls gene expression and protein synthesis. **FACT** Long-term memory needs new gene expression.
5	**Axon hillock** Trigger zone	Packed with voltage-gated Na^+ channels; the action potential starts here. **FACT** Fires when membrane reaches threshold (~−55 mV).

#	Structure	Function · exam fact
6	**Axon** Conduction	Carries the action potential away from the soma. **FACT** Wider axons conduct faster; some reach ~1 m long.
7	**Myelin sheath** Conduction	Fatty glial wrapping: oligodendrocytes (CNS), Schwann cells (PNS). **FACT** Multiple sclerosis destroys CNS myelin, slowing conduction.
8	**Node of Ranvier** Conduction	Unmyelinated gaps dense with Na^+ channels where the signal regenerates. **FACT** Saltatory conduction: AP "jumps" node to node.
9	**Axon terminals** Output	Boutons full of vesicles that release neurotransmitter. **FACT** Ca^{2+} entry triggers vesicle fusion and release.
10	**Synapse** Output	Junction: presynaptic terminal, cleft, postsynaptic membrane. **FACT** Cleft is ~20 nm wide; transmitter diffuses across.

Personal MCB80x study chart — not an official HarvardX/edX publication.

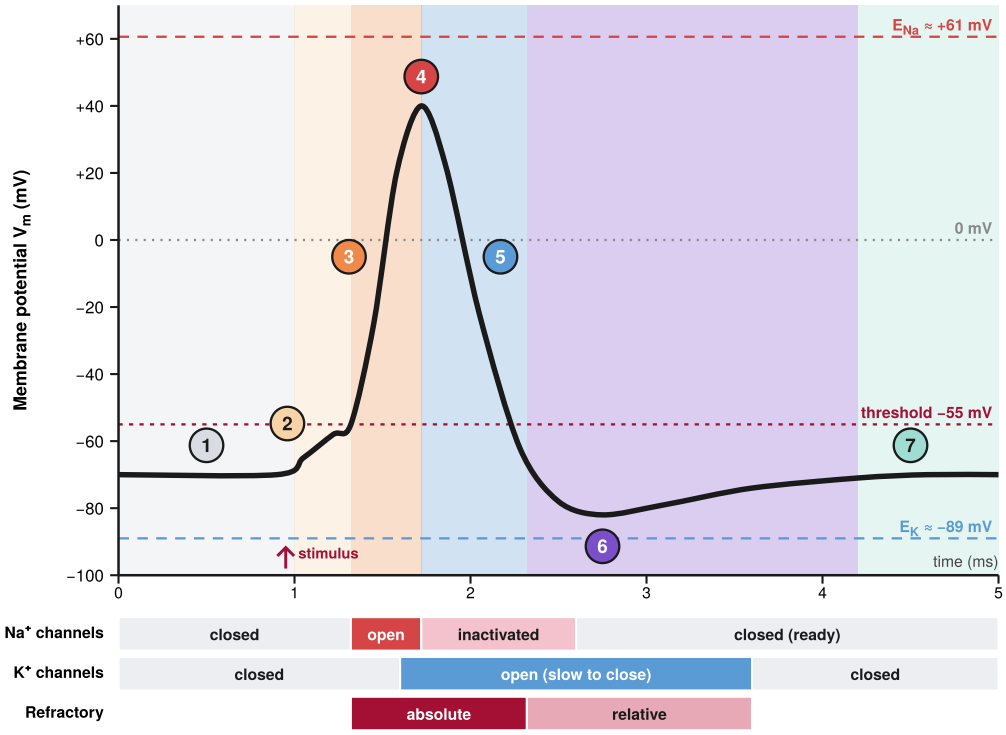
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

The Action Potential — Phases & Ion Channels

MCB80x · voltage-gated Na^+ and K^+ channels · refractory periods

ACTION POTENTIAL — V_m vs. TIME



Typical mammalian neuron at 37 °C. E values from your formula sheet's Example A.

PHASES

#	Phase	What happens · exam fact
1	**Resting potential** ≈ −70 mV	K^+ leak channels dominate, so V_m sits near E_K. **FACT** Rest is set mainly by K^+ permeability (Goldman equation).
2	**Depolarization** to threshold ≈ −55 mV	Stimulus or summed EPSPs push V_m upward. **FACT** Below threshold → no spike. All-or-none.
3	**Rising phase** Na^+ in	Voltage-gated Na^+ channels open; Na^+ rushes in, opening more channels (positive feedback). **FACT** Tetrodotoxin (TTX) blocks these Na^+ channels.
4	**Peak / overshoot** ≈ +40 mV	Na^+ channels inactivate; V_m heads toward but never reaches E_{Na}. **FACT** Inactivation gate = "ball-and-chain" plug.
5	**Falling phase** K^+ out	Delayed voltage-gated K^+ channels open; K^+ flows out and repolarizes. **FACT** TEA blocks K^+ channels → spike stays prolonged.
6	**Undershoot** ≈ −80 mV	K^+ channels close slowly, so V_m dips toward E_K. **FACT** Also called after-hyperpolarization.
7	**Return to rest** ≈ −70 mV	K^+ channels close; leak channels restore resting potential. **FACT** Na^+/K^+ pump keeps gradients long-term; not needed per spike.

Refractory periods

- **Absolute:** no new spike at any strength — Na^+ channels are open or inactivated.
- **Relative:** a stronger stimulus can fire — K^+ still open, some Na^+ recovered.
- Refractoriness makes spikes travel **one way** and caps firing rate.

Personal MCB80x study chart — not an official HarvardX/edX publication.

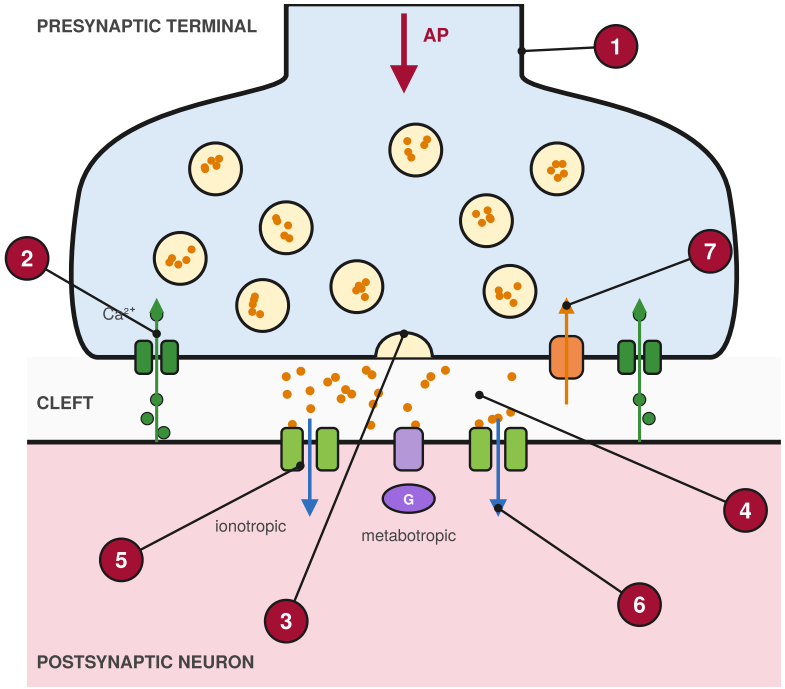
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

The Synapse & Neurotransmitters

MCB80x · chemical transmission step by step · major transmitters

CHEMICAL SYNAPSE



7 STEPS OF TRANSMISSION

1. **Action potential arrives.** Depolarizes the presynaptic terminal.
2. **Ca^{2+} channels open.** Voltage-gated Ca^{2+} channels let Ca^{2+} in — the key trigger.
3. **Vesicles fuse.** SNARE proteins dock vesicles; exocytosis releases transmitter.
4. **Diffusion.** Transmitter crosses the cleft in well under a millisecond.
5. **Receptor binding. Ionotropic** = ligand-gated channel (fast). **Metabotropic** = G-protein cascade (slow, longer).
6. **Postsynaptic potential. EPSP**: Na^+ in, depolarizes. **IPSP**: Cl^- in or K^+ out, hyperpolarizes.
7. **Clearance.** Reuptake transporters, enzymes (AChE), or diffusion end the signal.

MAJOR NEUROTRANSMITTERS

	Transmitter	Role · exam fact
Glu	**Glutamate** Amino acid · excitatory	Main excitatory transmitter in the CNS; AMPA and NMDA receptors. **FACT** NMDA needs glutamate *and* depolarization (Mg^{2+} block) → LTP.
GABA	**GABA** Amino acid · inhibitory	Main inhibitory transmitter in the brain; $GABA_A$ opens Cl^- channels. **FACT** Benzodiazepines and alcohol enhance $GABA_A$.
Gly	**Glycine** Amino acid · inhibitory	Inhibition in spinal cord and brainstem. **FACT** Strychnine blocks glycine receptors → convulsions.
ACh	**Acetylcholine** Cholinergic	Neuromuscular junction, autonomic system, attention and memory. **FACT** Cleared by AChE. Myasthenia gravis attacks nicotinic ACh receptors.
DA	**Dopamine** Monoamine	Reward, motivation, movement (substantia nigra). **FACT** Parkinson's = DA neuron loss. Cocaine blocks DA reuptake.
5-HT	**Serotonin** Monoamine	Mood, sleep, appetite; raphe nuclei. **FACT** SSRIs (e.g., fluoxetine) block serotonin reuptake.
NE	**Norepinephrine** Monoamine	Arousal, attention, fight-or-flight; locus coeruleus. **FACT** Also a sympathetic nervous system transmitter.

Rules of thumb

- Whether a transmitter excites or inhibits depends on the **receptor**, not the molecule (ACh excites skeletal muscle, slows the heart).
- One AP ≈ one small PSP; the cell fires only when PSPs **summate** to threshold at the hillock.

Personal MCB80x study chart — not an official HarvardX/edX publication.

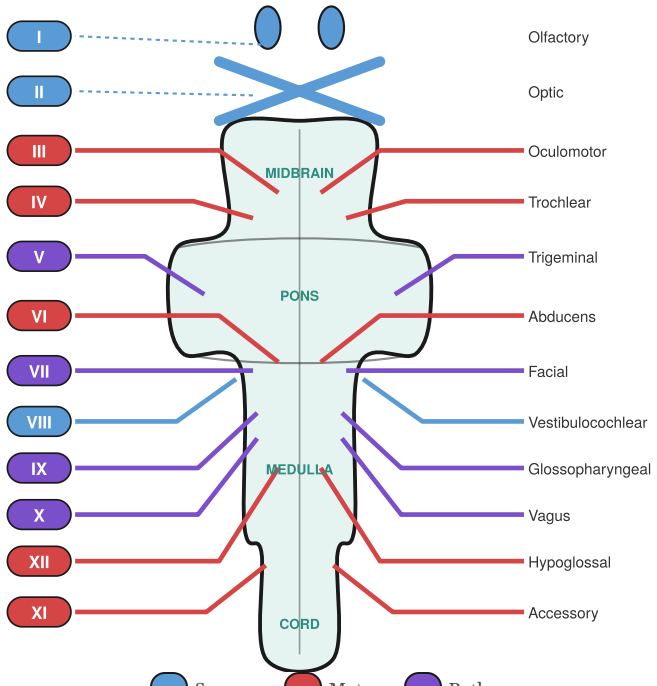
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Cranial Nerves I–XII

MCB80x · name, type, exit level, function, and exam facts

VENTRAL VIEW — WHERE EACH NERVE EXITS



Schematic; nerves are paired (left and right).

Mnemonics

- **Names:** *Ooh, Ooh, Ooh, To Touch And Feel Very Good Velvet, Ah Heaven.*
- **Type:** *Some Say Marry Money, But My Brother Says Big Brains Matter More.* (S = sensory, M = motor, B = both)
- **Eye muscles:** LR_6SO_4, all the rest by III.

THE 12 CRANIAL NERVES

#	Nerve · exit	Type	Function · exam fact
I	**Olfactory** Forebrain	Sensory	Smell. **FACT** Only sense that skips the thalamus on the way to cortex.
II	**Optic** Forebrain	Sensory	Vision. **FACT** Nasal fibers cross at the optic chiasm.
III	**Oculomotor** Midbrain	Motor	Most eye movements, eyelid lift, pupil constriction. **FACT** Damage: eye “down and out,” dilated pupil.
IV	**Trochlear** Midbrain	Motor	Superior oblique: moves eye down and in. **FACT** Only nerve exiting the back of the brainstem.
V	**Trigeminal** Pons	Both	Face sensation; chewing muscles. **FACT** Trigeminal neuralgia: stabbing facial pain.
VI	**Abducens** Pons–medulla junction	Motor	Lateral rectus: moves eye outward. **FACT** Damage: eye can’t look to the side.
VII	**Facial** Pons–medulla junction	Both	Facial expression; taste (front ⅔ of tongue); tears, saliva. **FACT** Bell’s palsy: one side of face droops.
VIII	**Vestibulocochlear** Pons–medulla junction	Sensory	Hearing and balance. **FACT** Acoustic neuroma → hearing loss, vertigo.
IX	**Glossopharyngeal** Medulla	Both	Taste (back ⅓); swallowing; carotid body sensing. **FACT** Gag reflex: IX sensory, X motor.
X	**Vagus** Medulla	Both	Parasympathetic to heart, lungs, gut; voice box. **FACT** “Wanders” all the way to the abdomen.
XI	**Accessory** Medulla + cervical cord	Motor	Trapezius and sternocleidomastoid: shrug, turn head. **FACT** Roots rise from the upper spinal cord.
XII	**Hypoglossal** Medulla	Motor	Tongue movement. **FACT** Damage: tongue deviates toward the injured side.

Personal MCB80x study chart — not an official HarvardX/edX publication.

Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Sensory Pathways — Vision & Hearing

MCB80x · receptor → brainstem → thalamus → cortex · visual field lesions

VISUAL PATHWAY — EYE TO CORTEX

Receptor · Nerve / tract · Brainstem · Thalamus · Cortex

Retina	Optic nerve	Optic chiasm	Optic tract	LGN	Optic radiations	V1 → streams
Eye Rods (dim light) and cones (color) → bipolar → ganglion cells.	Cranial nerve II Ganglion cell axons leave the eye; blind spot where they exit.	Below hypothalamus **Nasal** fibers cross; temporal fibers stay on the same side.	Each carries one visual field Right tract = left visual field from both eyes.	Thalamus 6 layers; keeps eyes and M/P streams separate.	White matter Fan out through parietal and temporal lobes to V1.	Occipital lobe **Ventral** "what" → temporal. **Dorsal** "where/how" → parietal.

AUDITORY PATHWAY — EAR TO CORTEX

Cochlea	Auditory nerve	Cochlear nuclei	Superior olive	Inferior colliculus	MGN	A1
Inner ear Hair cells bend → transduce sound. Basilar membrane maps frequency.	Cranial nerve VIII Carries frequency (place) and timing information.	Medulla First central stop; still one ear only.	Pons **First binaural** stop: compares the two ears to localize sound.	Midbrain Integrates all brainstem auditory input.	Thalamus Medial geniculate relays to cortex.	Temporal lobe Tonotopic map on Heschl's gyrus.

VISUAL FIELD LESIONS (black = lost field)

Lesion site	L eye · R eye	Deficit
Right optic nerve		Blind in the right eye.
Optic chiasm (center)		**Bitemporal hemianopia**, often from a pituitary tumor.
Right optic tract		**Left homonymous hemianopia** (same half lost in both eyes).
Right V1		Left homonymous hemianopia, often with **macular sparing**.

PRINCIPLES TO KNOW

- **Thalamic relay:** vision (LGN) and hearing (MGN) both pass the thalamus; only smell skips it.
- **Maps are kept:** retinotopy in vision, tonotopy in hearing, all the way to cortex.
- **Vision crosses at the chiasm**, so each hemisphere sees the opposite visual field.
- **Hearing is bilateral above the cochlear nuclei**, so one-sided cortex damage rarely deafens an ear.
- **Superior colliculus** (midbrain) also gets retinal input for reflexive eye and head orienting.

Personal MCB80x study chart — not an official HarvardX/edX publication.